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Roll No.....

Degree: B.Tech. Semester: III - Course work
MID-SEMESTER EXAMINATION, SEPTEMBER 2025

Course Title: Mathematics for Signal Processing and Communication
Course Code: ECMT C04

Time: 1:30 Hours

Max. Marks: 25

Note: Attempt all questions in given order only. Missing data/ information (if any), maybe suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Show that the set $H = \{A \in M_2(\mathbb{R}) : \text{trace}(A) = 0\}$ is a subgroup of a group $M_2(\mathbb{R})$, the set of all 2×2 matrices over $\mathbb{R}$.	2.5	CO1
1b	Define the term order of a group and the order of an element in a group. Illustrate with an example.	2.5	CO1
2a	Define cyclic group. Is $U(16)$ a cyclic group?	2.5	CO1
2b	Define homomorphism of a group. Show that the factor group $\frac{\mathbb{Z}}{4\mathbb{Z}}$ is isomorphic to $\mathbb{Z}_4$.	2.5	CO1
3a	Find all zero divisors in $\mathbb{Z}_6$. Is every field an integral domain? Justify your answer.	2.5	CO1
3b	Determine whether the set $W = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$ forms a subspace of $\mathbb{R}^3$.	2.5	CO2
4a	Investigate the linear independence or dependence of the set of vectors $x_1 = (1, 0, 0)$, $x_2 = (1, 1, 1)$, $x_3 = (0, 0, 1)$, $x_4 = (0, 1, 0)$ in $\mathbb{R}^3$.	2.5	CO2
4b	Consider the vector space $\mathbb{P}_2$ of polynomials of degree at most 2. Is the set $\{x, x + x^2, 1 + x^2\}$ forms a spanning set for the vector space $\mathbb{P}_2$? Justify your answer.	2.5	CO2
5a	Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(a_1, a_2) = (2a_1 + a_2, a_1 - a_2)$. Show that T is a linear transformation.	2.5	CO2
5b	Using Rank Nullity Theorem show that the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ given by $T(x, y, z) = (2x - y, y - z, z - x, 0)$ is not bijective.	2.5	CO2